

Followup 26-01

Drowning Cardiac Arrest

PATIENT	33-year-old male, no significant known history other than asthma
OUTCOME	Died following transition to comfort care on hospital day 2

The call

Dispatched for cardiac arrest at a public beach. Law enforcement and lifeguards reported the patient had been out of his friend's sight for roughly ten minutes before being found unresponsive in the water. Bystander CPR was in progress on arrival. Patient was moved to the roadway for EMS.

Field course:

- LUCAS applied. Pads placed, initial rhythm asystole. No shock advised at any point.
- i-gel attempted, would not seat. Large volume of sputum and water in the airway. Airway suctioned.
- Endotracheal intubation with video laryngoscopy and bougie. Confirmed by lung sounds, chest rise, and waveform capnography.
- IO access established.
- Four rounds of epinephrine, four cycles of ACLS.
- Crew departed scene early due to a large and growing crowd.
- ROSC achieved en route, approximately 28 minutes after the first documented crew entry. Post-ROSC rhythm sinus tachycardia with a strong pulse.
- **ETCO2 on arrival at the ED: 99 mmHg.**

Filed impression

Cardiac arrest secondary to drowning. Correct.

Hospital course

Emergency department. ACLS continued briefly on arrival. Cardiac activity confirmed on ultrasound. Pupils fixed and dilated, no gag, GCS 3. Hypotensive, requiring norepinephrine to maximum dose plus vasopressin. Right internal jugular central line placed. CT head showed diffuse cerebral edema with crowded sulci and small ventricles. CT of chest, abdomen, and pelvis showed nondisplaced fractures of the right costochondral cartilage at ribs 6 and 7, and no other acute injury. Cervical spine CT was negative for fracture.

Seizure-like activity was observed. Loaded with levetiracetam. Given mannitol. Intubation continued on mechanical ventilation. Total critical care time in the ED was 90 minutes.

Hospital day 0. Neurosurgery took him emergently to the operating room for a bifrontal decompressive craniectomy to relieve the cerebral edema. Estimated blood loss 60 mL. Documented prognosis at the close of the operative note: poor. Admitted to ICU for post-ROSC management, sedated on propofol.

Hospital day 1. Developed obstructive hydrocephalus at the fourth ventricle from cerebellar swelling. Returned to the OR for external ventricular drain placement.

Hospital day 2. Repeat CT showed cerebral edema persisting in the basal cisterns with herniation through the craniectomy defects, and at the medial temporal lobes and cerebellar tonsils. Neurology performed a brain death evaluation. Off sedation, he had no central or peripheral pain response in the upper extremities, absent pupillary, corneal, cough, gag, oculocephalic, and vestibulo-ocular reflexes.

He was **not** declared brain dead. Extension of the toes on lower extremity stimulation, and a nuclear medicine brain flow scan that demonstrated cerebral blood flow, both preclude the diagnosis. Prognosis remained catastrophic. Family transitioned him to comfort care.

Final diagnoses: Drowning and submersion in natural water. Cardiac arrest. Anoxic brain injury with malignant cerebral edema. High anion gap metabolic acidosis. Acute kidney injury. Hyponatremia.

Education points

Why the brain swelled

There are two ways a brain swells, and they are not the same problem.

Vasogenic edema is a leak. The blood brain barrier breaks down and fluid escapes from the vessels into the tissue around them. This is what you see around tumors, abscesses, and some contusions. It responds reasonably well to steroids.

Cytotoxic edema is what killed this patient, and it happens inside the cells. Every neuron runs a pump in its membrane called the sodium-potassium ATPase, which continuously drags sodium out of the cell. That pump is the single largest consumer of energy in the brain, and it runs on ATP. ATP is made with oxygen. When oxygen stops, ATP production stops within a couple of minutes, the pump quits, and sodium floods into the cell. Water follows sodium. Every affected cell in the brain takes on water and swells at once.

Note what that means. The swelling is not a reaction to the injury that develops over hours. It is the injury, and it begins at the moment perfusion stops. Steroids do nothing for it. The neurosurgeon's preoperative diagnosis on this case was written exactly that way: global cytotoxic cerebral edema.

Why swelling is uniquely lethal in the head

The skull is a sealed rigid box with a fixed internal volume. It holds three things: brain tissue, blood, and cerebrospinal fluid. Because the box cannot expand, any increase in one of those three has to be matched by a decrease in another. This is the **Monro-Kellie doctrine**, and it is the most important concept in neurologic emergency care.

The brain compensates at first by squeezing CSF out of the ventricles and cisterns and pushing venous blood out of the cranium. That is why his first CT was read as crowded sulci and small ventricles. Those spaces were being

consumed. The compensation is finite. Once the CSF and venous reserve are spent, further swelling produces a steep and accelerating rise in intracranial pressure.

Then the pressure itself becomes the injury. Perfusion pressure to the brain is mean arterial pressure minus intracranial pressure. As ICP climbs, perfusion falls, which starves more neurons, which causes more cytotoxic swelling, which raises ICP further. It is a closed loop that feeds itself, and it is why a patient can be resuscitated successfully and still lose the brain over the next 48 hours.

The anatomy of herniation

Inside the skull, two tough sheets of dura divide the space into compartments. The **falx cerebri** runs front to back in the midline and separates the two cerebral hemispheres. The **tentorium cerebelli** is a horizontal shelf that separates the cerebrum above from the cerebellum and brainstem below, with an opening in it called the tentorial notch. Below all of that, the **foramen magnum** is the hole at the base of the skull where the brainstem becomes the spinal cord.

When pressure exceeds what a compartment can hold, brain tissue is forced through the nearest opening. That is herniation, and the route determines what dies:

Subfalcine. Tissue slides under the falx across the midline. Often the earliest and least immediately fatal.

Transtentorial, also called uncal. The medial temporal lobe is pushed down through the tentorial notch. It compresses the third cranial nerve, which is why a fixed and dilated pupil on one side is the classic sign, and then compresses the midbrain itself.

Tonsillar. The cerebellar tonsils are forced down through the foramen magnum and crush the medulla. The medulla contains the respiratory and cardiovascular control centers. This one kills quickly.

This patient's day 2 CT documented **both** medial temporal and tonsillar herniation. By that point the injury had reached the brainstem.

What a craniectomy is

The distinction is in the suffix and it matters.

A **craniotomy** removes a piece of skull to reach the brain, and the bone flap is replaced and secured at the end of the operation.

A **craniectomy** removes the bone and leaves it off. That is the entire therapeutic point. Removing bone converts the sealed box into an open one, giving the swelling somewhere to expand other than through the tentorium and foramen magnum. It does not treat the edema. It buys space and time while the edema runs its course.

Bifrontal means the bone was taken from both frontal regions as one large piece spanning the midline, rather than from one side. It is also called a Kjellberg craniectomy. The approach is a bicoronal incision, a curved cut across the top of the scalp running roughly ear to ear, with the scalp and muscle folded forward as a flap.

Two details in this operative note are worth understanding. The surgeon **ligated and divided the superior sagittal sinus anteriorly**, and **released the falx**. The superior sagittal sinus is a large venous drainage channel that runs inside the top edge of the falx. Its anterior third can be sacrificed without catastrophic venous congestion, while the posterior portion cannot, so this is a deliberate and anatomically bounded maneuver. Cutting the sinus and the falx anteriorly unhooks the midline tether so both hemispheres can expand forward together into the space the bone removal created. Without that release, a bifrontal decompression is far less effective.

The EVD, and why he needed one

CSF is produced continuously by the choroid plexus inside the ventricles and follows a fixed path: the two lateral ventricles drain through the foramina of Monro into the third ventricle, then down the narrow cerebral aqueduct into the fourth ventricle, then out into the space around the brain and spinal cord where it is reabsorbed.

That path has choke points. On hospital day 1, his cerebellum had swollen enough to compress and efface the fourth ventricle. CSF could still be made but could no longer get out. Pressure backed up upstream and the ventricles dilated. This is **obstructive hydrocephalus**, and it is a second and separate pressure problem stacked on top of the first.

An **external ventricular drain** is a catheter passed through the skull directly into a lateral ventricle. It does two jobs at once: it drains CSF to relieve the backup, and because it sits in a fluid column, it measures intracranial pressure directly.

Why none of it worked

The decompression and the drain both did what they were designed to do. The repeat imaging confirmed the ventriculomegaly had resolved. The edema kept progressing anyway, and the brain herniated through the craniectomy defects themselves.

Surgery can manage pressure. It cannot restore neurons that died from anoxia, and it cannot stop a cytotoxic process that was set in motion before anyone reached him. Every one of these interventions was aimed at the consequences of the injury. The injury was complete underwater.

What this means in the truck

Everything above happens downstream of three variables you control directly.

Oxygenation. The cytotoxic cascade starts with an ATP failure that starts with an absence of oxygen. Hypoxia after ROSC continues the injury.

Blood pressure. Perfusion pressure is MAP minus ICP. When ICP is rising, a soft MAP is not benign. Post-ROSC hypotension is second insult to a brain already failing.

Ventilation. CO₂ is the most potent cerebral vasodilator in the body. Hypercarbia dilates cerebral vessels, increases intracranial blood volume, and raises ICP in a box that has no room left. Aggressive hyperventilation is not the answer either, because it constricts those vessels and cuts perfusion to tissue that is already starving. The target is normal, and the only way to know you are there is a waveform capnography number you are actually watching.

The labs, and what they mean

Values are grouped by the story they tell, not by the panel they came on. Bolded values are the ones that carry the teaching. Reference ranges are the reporting laboratory's own where the source report supplied one, and standard adult values where it did not. Ranges vary between institutions and between analyzers, so use the range printed on the report in front of you rather than memorizing these.

Acid and lactate: the size of the oxygen debt

Lab	Reference range	Arrival	2 hr	4 hr	Day 2
Lactic acid	0.0 to 2.0 mmol/L	26.6	11.2	3.8	7.5
pH	7.35 to 7.45	7.07	7.13	7.13	6.96
PaCO ₂	35 to 48 mmHg	64	57	51	86
Bicarbonate	22 to 29 mmol/L	7		19	13
Anion gap	5 to 14 mmol/L	35			28
Base excess	-2.0 to 3.0 mmol/L	-12.4	-11.0	-12.5	-13.8
PaO ₂	83 to 108 mmHg	154	340	135	99

Lactate is what cells produce when they make energy without oxygen. It is a direct readout of how long and how badly tissue went without perfusion. Normal is under 2. This patient arrived at **26.6**, which is among the highest values you will ever see in a living patient. That number is the submersion time and the arrest time added together and converted into chemistry.

Anion gap is the arithmetic that tells you an unmeasured acid is present. Sodium minus chloride minus bicarbonate. A gap of 35 with a lactate of 26.6 means essentially the entire gap was lactate.

Bicarbonate of 7 means the body's buffer was nearly spent. Bicarbonate is what neutralizes acid, and it was consumed defending against the lactate.

The gas is a **mixed acidosis**. The pH of 7.07 is not just metabolic. The PaCO₂ of 64 means he was also not clearing CO₂, so a respiratory acidosis was stacked on top of a metabolic one. Two problems pushing pH the same direction.

Read the trend across the first four hours: lactate falls 26.6 to 11.2 to 3.8, PaCO₂ falls 64 to 57 to 51. That is what restored perfusion and effective ventilation look like on paper. Perfusion returned and the debt got paid down. Then look at day 2: lactate climbs back to 7.5, PaCO₂ to 86, pH to 6.96. The organs were failing by then and the direction reversed.

On your ETCO₂ of 99. ETCO₂ is a perfusion monitor as much as a ventilation monitor. During arrest it is low because there is no cardiac output to carry CO₂ to the lungs. At ROSC, circulation suddenly returns and delivers a backlog of CO₂ that has been accumulating in underperfused tissue for the entire arrest. That washout surge is why ETCO₂ spikes high immediately post-ROSC in a long down time. **Your 99 was corroborated by the ED arterial gas.** It was real, and it was one of the earliest quantitative signs of how deep the oxygen debt was. A sudden ETCO₂ rise during CPR is one of the most reliable field indicators of ROSC.

Kidney and muscle: rhabdomyolysis

Lab	Reference range	Arrival	Day 2
Creatinine	0.70 to 1.20 mg/dL	1.35	8.07
BUN	7 to 20 mg/dL		69
eGFR	over 60 mL/min/1.73 m ²		8
Creatine kinase	20 to 200 U/L	10,048	12,047
Potassium	3.5 to 5.1 mmol/L		5.9
Phosphorus	2.5 to 4.5 mg/dL		9.1
Calcium	8.6 to 10.4 mg/dL	7.7	8.2

Creatinine measures kidney filtration. It went from nearly normal on arrival to eight over 48 hours. That climb is acute kidney injury from two sources at once: the kidneys were underperfused during the arrest, and they were then clogged by the products of dying muscle.

Creatine kinase is an enzyme released when skeletal muscle breaks down. Over 10,000 on arrival, still climbing on day 2. This is **rhabdomyolysis**. Muscle that sat without blood flow during the arrest dies and dumps its contents into the circulation, including myoglobin, which is directly toxic to the renal tubules.

Here is a detail worth memorizing. His urine dipstick was **moderate positive for blood**, but the microscope found only **1 red blood cell**. Blood on the dipstick with no red cells under the scope is not blood. It is **myoglobin**, which cross-reacts on the same dipstick pad. That mismatch is a classic and near-diagnostic sign of rhabdomyolysis.

Potassium of 5.9 and phosphorus of 9.1 are the other half of that picture. Both live inside muscle cells. When muscle lyses, they spill out. Add failing kidneys that cannot excrete them and the potassium becomes a cardiac threat in its own right.

Liver: shock liver

Lab	Reference range	Arrival	Day 2
AST	0 to 50 U/L	59	over 7,000
ALT	0 to 50 U/L	69	4,413
Total bilirubin	0.2 to 1.3 mg/dL	under 0.2	0.5
Prothrombin time	9.5 to 12.0 seconds	16.8	
INR	0.9 to 1.2	1.6	
Total protein	6.4 to 8.3 g/dL	6.1	5.7
Albumin	3.5 to 5.0 g/dL		3.1

AST and ALT are enzymes inside liver cells. When those cells die, the enzymes leak into the blood. On arrival they were barely elevated. Within 48 hours they were in the thousands.

This is **ischemic hepatitis**, commonly called shock liver. It is the signature of a liver that was starved of blood flow. Note the pattern: transaminases in the thousands with a **normal bilirubin**. That combination distinguishes shock liver from most other causes of liver injury, where bilirubin usually rises alongside. The enzyme rise is also delayed. His arrival values looked almost reassuring. The damage was already done and simply had not shown up in the blood yet.

INR of 1.6 on the first night is the same injury from a different angle. The liver manufactures clotting factors. A liver that is dying makes fewer of them, and the blood takes longer to clot.

Heart: it was never the problem

Lab	Reference range	Value
hsTnT baseline	under 23 ng/L (male)	13
hsTnT at 2 hr	under 23 ng/L; critical at 100 or above	138 , flagged critical
Delta from baseline	under 8 ng/L	125
Ejection fraction	55 to 70 percent	70 to 75 percent

Troponin is released when heart muscle is injured. His rose from 13 to 138 within two hours.

Do not read that as a heart attack. Troponin rises whenever cardiac muscle is stressed, and a prolonged arrest with epinephrine, chest compressions, and profound hypoxia does exactly that. This is **type 2 injury**, meaning oxygen demand outstripped supply. It is not a blocked coronary artery.

The echocardiogram settles it. **Ejection fraction 70 to 75 percent**, which is normal to hyperdynamic, with no valve disease. His heart was structurally fine.

That is the point of this section. **A drowning arrest is a respiratory arrest**. The heart stopped because it ran out of oxygen, not because of a primary cardiac event. This is why the initial rhythm was asystole and no shock was ever advised, and it is why airway and ventilation are the priority in drowning rather than the defibrillator. His heart did not fail him. His airway did, and then his brain paid for it.

Sodium and the brain

Lab	Reference range	Early	Day 2
Sodium	136 to 145 mmol/L	126 to 132	148, then 152
Chloride	98 to 107 mmol/L	97	111
Serum osmolality	270 to 300 mOsmol/kg	308 to 316	328

Sodium ran **126 to 132** early, and then was deliberately driven up to **148 and then 152** by day 2.

Sodium determines how water distributes between the blood and the tissues. Low sodium pulls water into cells, including brain cells, which worsens cerebral edema. That is why the ICU set an explicit target of 145 to 155 and considered hypertonic saline. In a patient with a swelling brain, a sodium at the low end is actively dangerous, and the goal is a controlled elevation to pull water back out of the brain.

The team also flagged **increased urine output** and asked to rule out central diabetes insipidus. Severe brain injury damages the pituitary's ability to release antidiuretic hormone. Without it, the kidneys dump free water, urine output becomes enormous, and serum sodium climbs on its own. In a head injury patient, high volume dilute urine is a red flag, not a sign of good kidney function.

Glucose and white count: stress, not disease

Lab	Reference range	Arrival	Later
Glucose	70 to 99 mg/dL	354	340, then 143
Hemoglobin A1c	under 5.7 percent		5.7
White blood cells	4.0 to 10.8 x10 ⁹ /L	10.9	29.7
Neutrophils	35.0 to 80.0 percent		82.8
Immature granulocytes	0.0 to 0.6 percent	6.4	3.5
Hemoglobin	14.0 to 18.0 g/dL	13.9	13.7
Platelets	150 to 400 x10 ⁹ /L	419	157

Glucose 354 on arrival, with **hemoglobin A1c 5.7**, which is at the top of normal. A1c reflects average glucose over roughly three months. His was fine, which means he was not diabetic. This was **stress hyperglycemia**, driven by the catecholamine and cortisol surge of an arrest and by four rounds of epinephrine.

White blood cell count 10.9 on arrival, 29.7 within a day, with a heavy neutrophil predominance and immature granulocytes. That looks like a raging infection and it was not. Massive tissue injury and catecholamine surge both demarginate white cells off the vessel walls and push the marrow to release immature forms. This is a **reactive leukocytosis**. He was covered with antibiotics as a precaution, correctly, but the count itself was the arrest talking.

Rib fractures

Nondisplaced fractures of the right costochondral cartilage at ribs 6 and 7. Anterior, at the junction of rib and cartilage, exactly where compressions land. This is an expected consequence of effective CPR, not a complication. Compressions that never break anything are frequently compressions that were too shallow.

Learning points

1. The pupil exam early after arrest does not tell you the outcome. His pupils were fixed and dilated in the ED, pinpoint a few hours later, and 6 mm and nonreactive on day 2. Sedation, hypoxia, epinephrine, and acidosis all move the pupils. A bad neuro exam in the first hours post-ROSC is expected and is not prognostic. Do not tell a family what a pupil means on the day of the arrest.

2. Airway in drowning is a suction problem before it is a tube problem. The i-gel would not seat because the mouth was full of water and sputum. That is not a device failure or an operator failure, it is the pathology. Supraglottic airways depend on a seal, and a seal cannot form against fluid. Suction first, aggressively, and expect to go to a definitive airway. The bougie and video laryngoscopy were the right escalation.

3. Brain death is a strict diagnosis, not a description. He had absent brainstem reflexes across the board and a devastating CT, and he still could not be declared brain dead. Any motor response, even reflexive extension of the toes, and any demonstrated cerebral blood flow, disqualifies the declaration. When you hear a family or a crew say someone is brain dead, it usually means something else. Know the difference.

4. Most of the damage from an arrest arrives after ROSC. Compare his arrival labs to day 2. Creatinine 1.35 to 8.07. AST 59 to over 7,000. Almost everything looked survivable on the first draw. The injury was already committed at the moment of arrest, and the labs simply had not caught up. This is why post-ROSC care is its own resuscitation and not a victory lap.

5. Return of circulation is not return of perfusion to everything. ROSC gets you a pulse. It does not undo the oxygen debt, and this case quantifies that debt precisely: a lactate of 26.6, a bicarbonate of 7, and a brain that swelled until it herniated. The clock that mattered started when he went under the water, and by the time anyone had hands on him, most of the outcome was already determined.

Prepared from hospital records obtained through the follow-up request process. De-identified for education. If you have questions about this case or want to request follow-up on one of your own, email abarker@medic1ambulance.org with FOLLOW-UP in the subject line.